Presentation 1

1. Link to presentation and code on GitHub:
    - Presentation Link: Click Here (Note: if the Power BI embedded in the slides fails, static visuals can be found at the bottom of the slide deck)
    - Coding was done on PowerBI. The data was originally transformed by my partner, Julia Schaffner, and then I built the visualization per the process below.

2. What did you do?

I created a visual for the Time Trends Dashboard using the CAR data. I built off my group partner, Julia's, work in which she compiled all CAR data and All Calls data into singular CSV files using the import code provided by Prof. Krishna. She then loaded them into PowerBI to perform more specific data transformations. Within Power BI, she used the Group By and Advanced Group By functions to ensure each Contact Session ID in the CAR dataset appeared only once, preventing double counting from multiple actions per session. Julia then created four new aggregated columns—Count, Call Start Time, Starting Hour, and All Rows—and expanded the nested tables to access detailed session information in the visuals.

With this data, I created a line chart visualization that aimed to examine if there was a seasonal peak in the number of calls received by Legal Aid. I made sure to filter the data so it only displays call volumes from 8/1/2024 - 7/31/2025 to ensure no double counting. The X-axis is the month in which the call took place over the course of this calendar year. The Y-axis is the number of unique calls during that month. On each visual, you can filter by EP Name, Flow Name, Termination Reason, and the axes for the visual.

3. How does it help the project?

Understanding when call volumes peak is crucial for improving the efficiency of the call system, especially given the constraints legal aid organizations face. With limited budgets and unpredictable funding due to shifting policies and economic conditions, Legal Aid is already struggling to meet demand. Calls often exceed capacity, leaving people unable to get through or facing long wait times. Since demand likely fluctuates rather than remaining constant—spiking during particular days, weeks, or policy changes—identifying these seasonal or situational trends allows the organization to anticipate and prepare for them. Though more analysis is needed, it currently looks like call volumes acutely spike in the new year, dip moderately during the summer months, and drop off dramatically in December. With this in mind, Legal Aid could leverage short-term staff or AI-powered agents during the new year while cutting back during the summer and the holidays. By aligning staffing and resources with periods of highest demand, the organization can serve more clients effectively without increasing costs, improving both accessibility and overall system performance.

4. Issues faced, attempt to resolve issues, and resolved issues:

The major issue I, along with the rest of my group, are facing is that the data has a lot of confusing and/or missing terms and information. We were able to get some context about acronyms and terms used by searching the internet or referencing the document Professor Arvind uploaded to GitHub; however, we still have questions about how the different answers in a column compare to each other in context of the overall business problem (i.e. SIP vs WXCC vs. WXC).

5. Next Steps

   - Drill down further to see if the call volume trend observed above is uniform across all menu types or if specific menu options follow unique trends and thus require different solutions/recommendations

- Compare if the seasonal spike from April to September is true in other years suggesting a regular trend or if it is more closely related to the extreme policy changes from this summer; to do so, I would create this same visualization with data from years past
- Refine the dashboard visuals to clearly display trends in peak call times by current filters; to do so, I would create heatmaps or bar graphs to highlight patterns across hours and days
- Add more filtering options to allow deeper analysis of when and why specific termination reasons happen
- Work closely with teammates analyzing other metrics to link time-trend findings with their insights for a more complete understanding

6. References

- Input code from Krish
- Julia’s data transformations in python and PowerBI
- Information from Cynthia during class with her

Presentation 2

1. Link to presentation and code on GitHub:
    - Presentation Link: Click Here (Note: static visuals used in presentation; to see live dashboard use link below)
    - Dashboard Link: Click Here
    - Data cleaning code: see Julia's code attached below
        - Click Here
        - Click Here

2. What did you do?

This week, my goal was to build off my previous visualization—which tracks call volumes by month—by allowing the viewer to see call volumes for each menu while adding hierarchical filtering by year and month. This way, Cynthia can easily compare how many callers are there for each menu across the relevant time frames that she needs. This visual is supposed to compliment Julia's visual as they seek to display similar information across different time frames; Julia displays each menu's usage over the course of a day whereas mine shows it over the course of a year.

To do so, I had to first download the new CAR data for the month of September. Then, using Krish's input code and Julia's data cleaning and hierarchical filtering code, I created three .csv files. One was the combined the raw data into a single .csv (using Krish's code); another had the data grouped by 'Contact Session ID' while also calculating and creating columns to track the month and year of the call; and the last create a hierarchical structure across five tiers so each activity can be traced back through its broader categories.

Next, I uploaded the three data sets to Power BI where I linked the first dataset to the second by 'Contact Session ID' and linked the third dataset to the first by 'Activity Name.' Here, I made a clustered bar graph to display the number of unique call session ID's associated with a menu option on a monthly basis. With this in mind, the x-axis was the months in the year whereas the y-axis was the number of distinct 'Call Session IDs.' The graph does display a drop off in the number of calls Legal Aid received so far in 2025 compared to 2024. However, my hypothesis presented in class—that certain menu options that are politically charged right now (e.g. immigration, HIV, etc.) may be driving the drop off—proved to be wrong. In fact, there appeared to be a relatively proportional drop off across all menu options, which makes me wonder why there just appears to be fewer people calling in. I hope to investigate this further in the coming weeks.

3. How does it help the project?

Understanding when call volumes peak is crucial for improving the efficiency of the call system, especially given the constraints legal aid organizations face. With limited budgets and unpredictable funding due to shifting policies and economic conditions, Legal Aid is already struggling to meet demand. Calls often exceed capacity, leaving people unable to get through or facing long wait times. Since demand likely fluctuates rather than remaining constant—spiking during particular days, weeks, or policy changes—identifying these seasonal or situational trends allows the organization to anticipate and prepare for them. Though more analysis is needed, it currently looks like call volumes on track to be less for 2025 compared to 2024. Additionally, it appears that call volumes across all menu options reduced proportionally between 2024 and 2025. We need to understand why this is occurring before making any concrete recommendations.

4. Issues faced, attempt to resolve issues, and resolved issues:

I didn’t face many significant issues during this revision. In the past weeks, I think Julia and I were not collaborating as closely as we should have been (especially given how our dashboards are meant to work in tandem). This was due mainly to us misunderstanding our objectives but we have corrected that this week and will continue to work closely moving forward.

5. Next Steps

   - Continue to investigate why call volumes dropped off between 2025 and 2024 across all menu types. I will dig through the data to see if there are other trends worth analysing to help answer this question.
   - Work closely with teammates analyzing other metrics to link time-trend findings with their insights for a more complete understanding

6. References

   - Input code from Krish
   - Julia’s data transformations in python
   - Information from Cynthia during class with her